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As such the blog is really good and can be submitted. I have some suggestions - but feel free to only include the ones you also agree with.

Can We Escape the Trade-Off Between Fast and Realistic Synthetic Anomalies?



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9 min read · Draft



Imagine you are building an anomaly detector for a factory.

You have thousands of images of perfectly normal products. What you do not have are thousands of examples of the things that can go wrong: a tiny foreign particle, a subtle contamination mark, a material fragment, a small surface defect.

And that creates a surprisingly awkward problem.

How do you train or calibrate an anomaly detector for defects that you barely have examples of?

One answer is to generate them synthetically.

But there is a catch.

Can you add an overview or some related image here; Something that shows synthetic anomaly generation or a lack of abnormal images in the anomaly detection.

If possible, please do not generate this image using AI, instead take it from some survey papers and cite or give credits to the paper at the end.

A production line might capture thousands of images of an intact component, while a particular manufacturing defect may occur only a handful of times. Some defects are also difficult to reproduce deliberately, making real data collection expensive and sometimes impractical.

Synthetic anomalies offer a way around this.

Here, you can also mention that another problem is that sometimes, the type of anomalies that can happen can not be foreseen before. What do you think

A procedural generator can take a normal image and perturb it with noise, texture, or geometric masks. It is fast and controllable, but its notion of a defect is often generic.

Generative models take the opposite approach. They can produce richer visual changes and condition the anomaly on the object and its context. The downside is computational cost.

This creates a familiar trade-off:

I guess here you can add a small explanation/simpler sentence to tell what these two types mean for a person who is not familiar with these words. Or even better would be to show an example using an image.

FLASH takes a different route.

It does not ask a generative model to create every final anomaly.

Instead, it uses generative modeling where it adds the most value: **creating realistic defect examples.**

Everything after that can be made substantially cheaper.

You could also add a sentence like "Flash uses the best of both worlds .."

that transfer to real defects.

And What Happens at the Pixel Level?

Image-level detection is only part of the story.

At the pixel level, FLASH provides the strongest synthetic result for **through the five detectors.**

More importantly, the per-category results reveal where synthetic anomalies still struggle.

This title needs to be at a lower title level (like, if this a H2, make it H3 ###) as this is a part of the "### Does It Actually Look Convincing?" section

What Does the Full Pipeline Look Like?

At this point, the individual components come together.

A normal host image enters the synthesis pipeline.

OBS identifies the object-aware region.

This section can be above the results sections and results ("## Does It Actually Look Convincing?" section) can be after this.

Explore the FLASH project

Link:

Do not forget to add the link.

Great write up. I like the flow and short sentences. Very easy to read and follow.